

Building Question Answering Models for Seismic Visualization

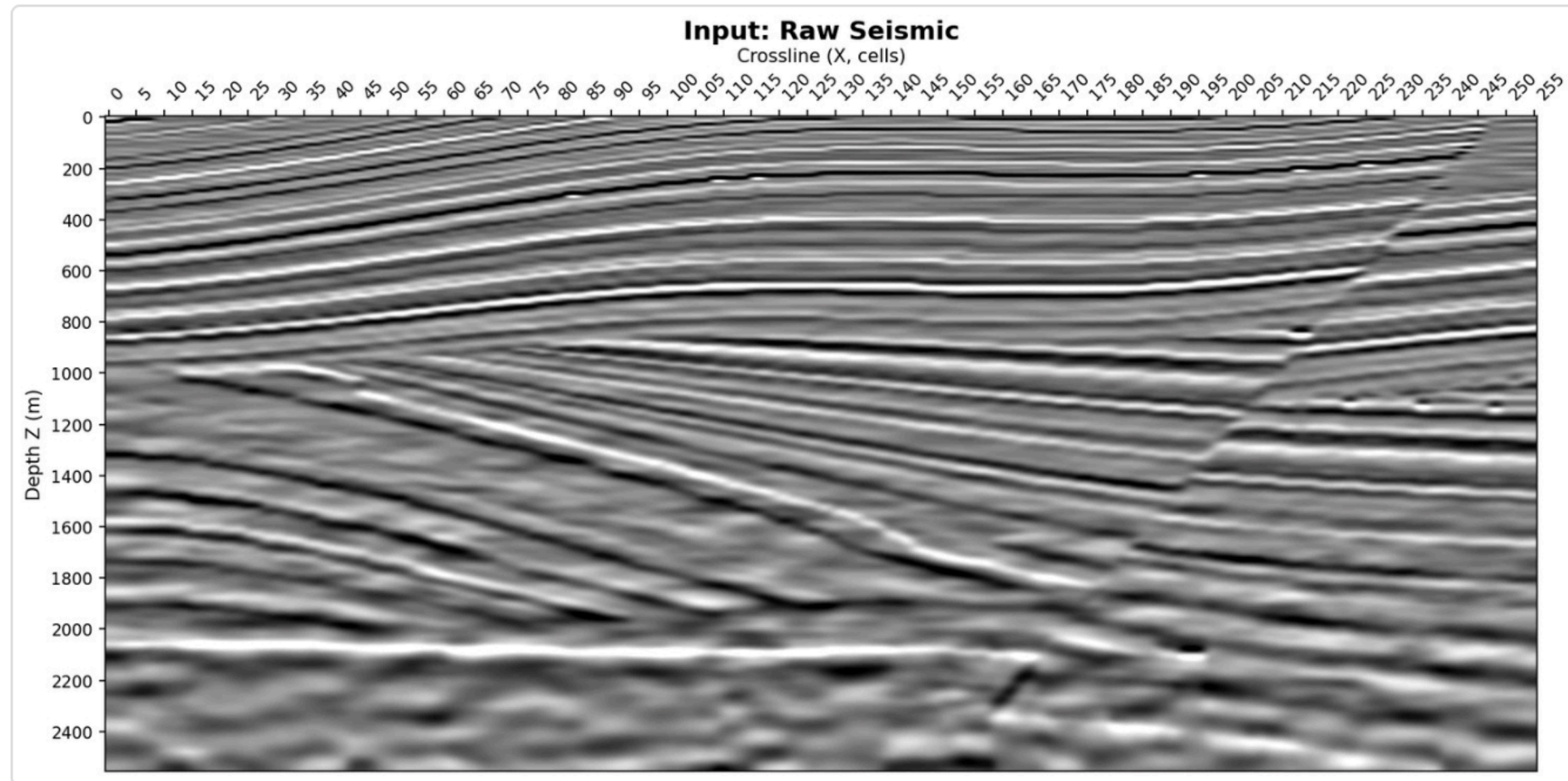


Fangcong Yin, Greg Durrett
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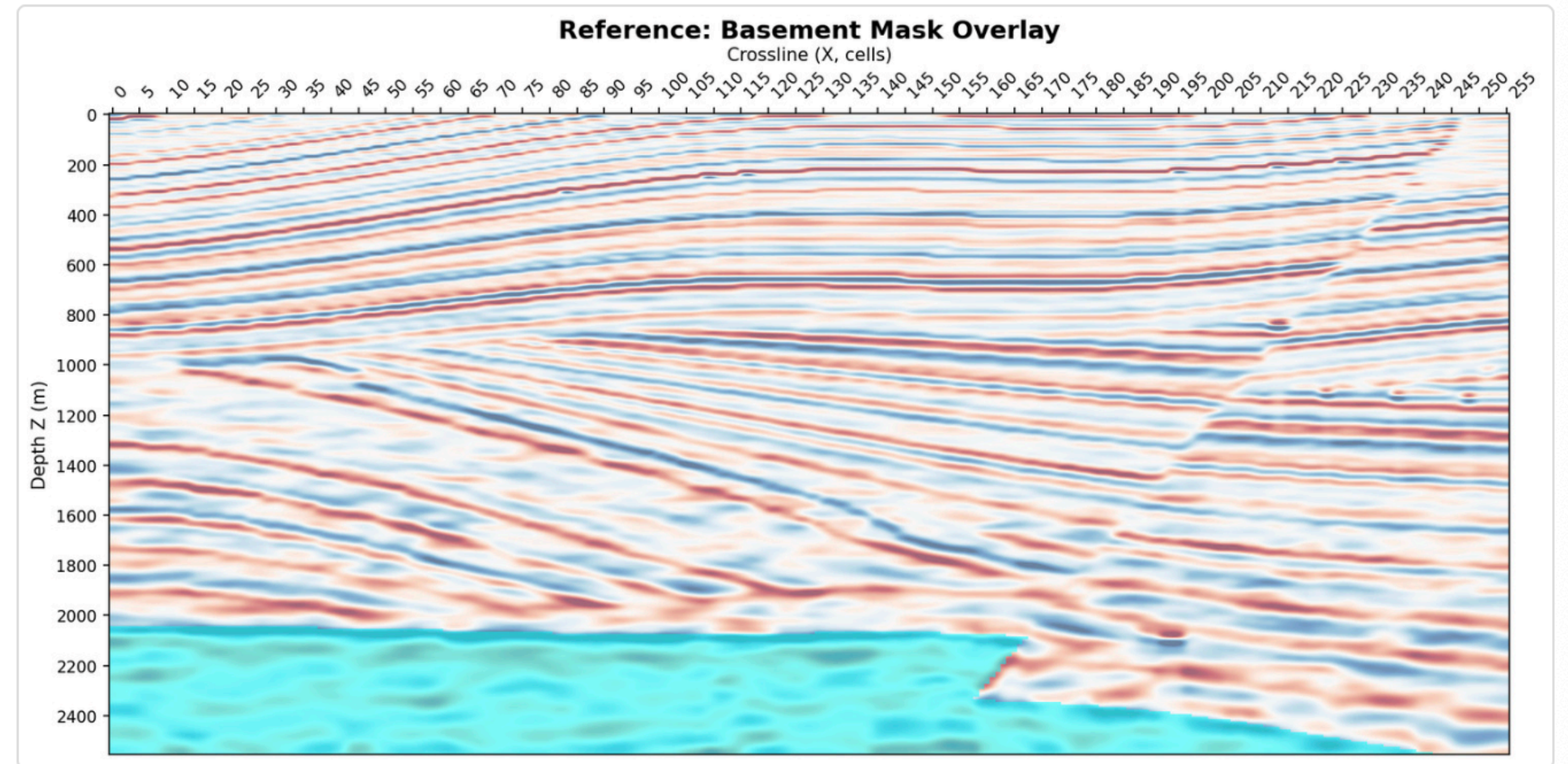


New Train Data

GRAYSCALE AGC PANEL (MODEL INPUT)



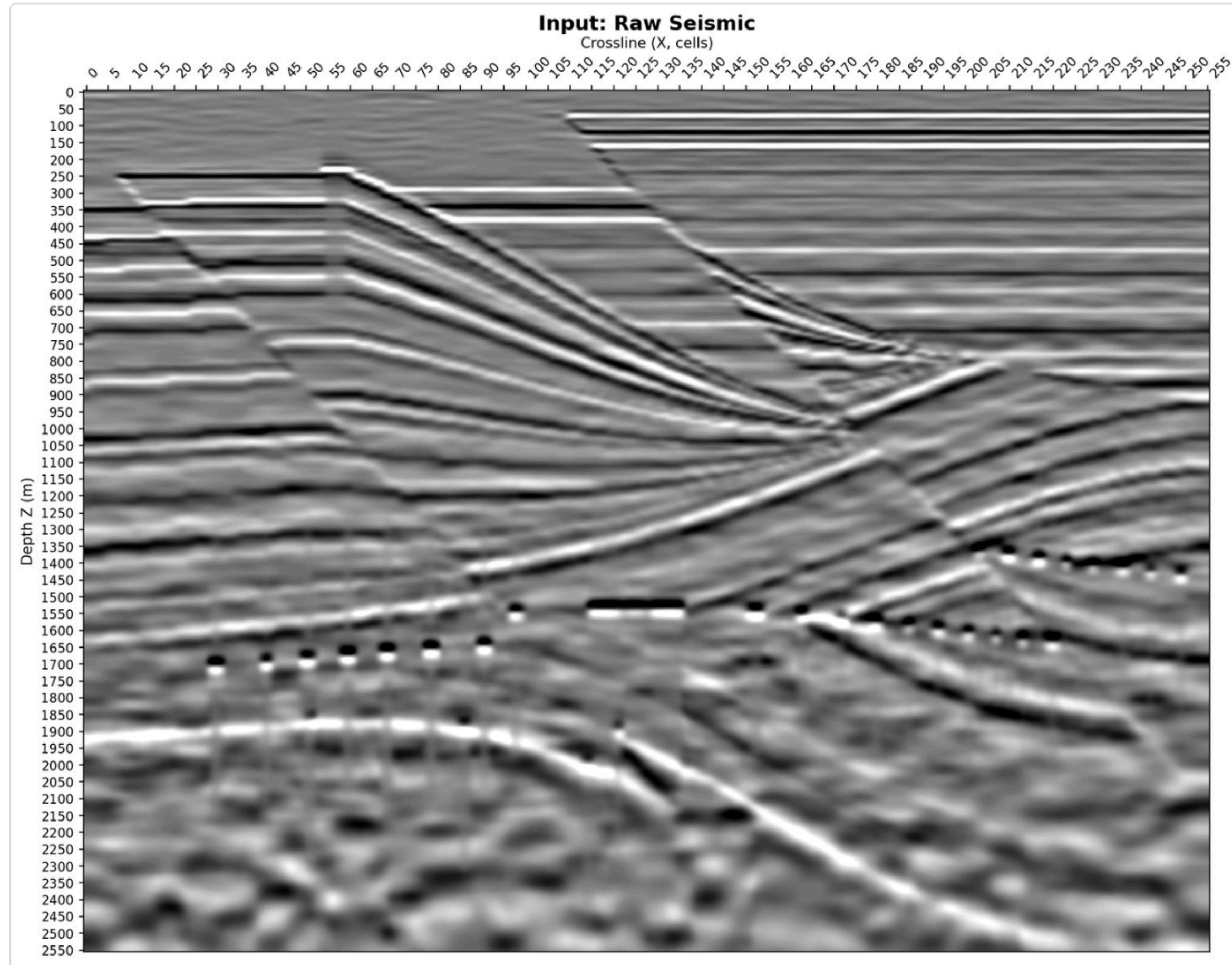
GT BASEMENT OVERLAY (REFERENCE ONLY)



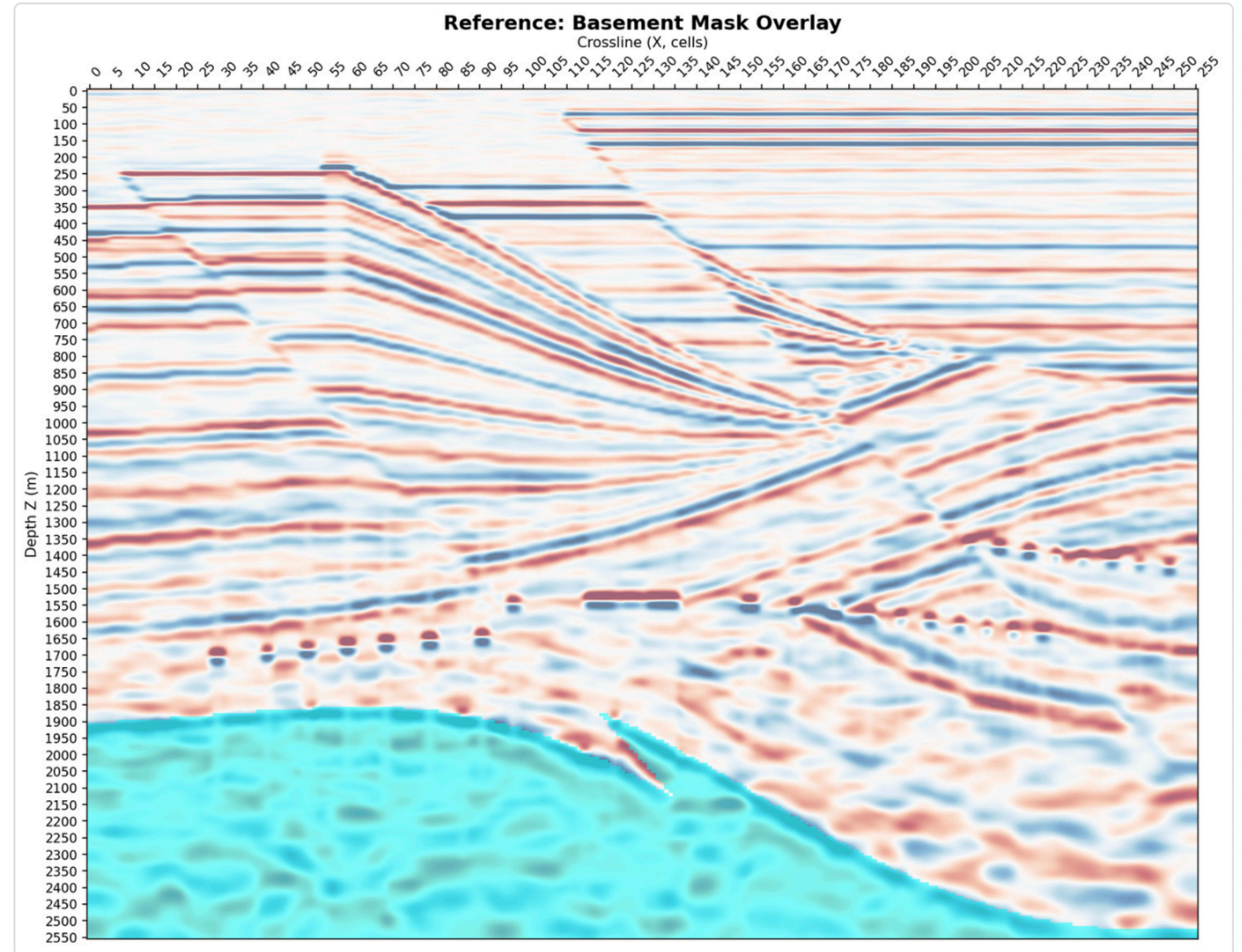


New Train Data

GRAYSCALE AGC PANEL (MODEL INPUT)



GT BASEMENT OVERLAY (REFERENCE ONLY)





Results

Bucket	n	pass	Accuracy ($\pm 25\text{m}$ for thickness)			Our model thickness acc within tolerance		
			Our model	Qwen2.5-VL-7B	gpt-5-mini	Ours $\pm 50\text{ m}$	Ours $\pm 100\text{ m}$	Ours $\pm 200\text{ m}$
Depth (thickness)	184	95	51.6%	13.0%	21.2%	60.9%	77.7%	91.3%
Thickest loc.	29	17	58.6%	34.5%	17.2%	—	—	—
Thickest loc. thickness	29	16	55.2%	17.2%	27.6%	55.2%	75.9%	93.1%
Thinnest loc.	29	15	51.7%	41.4%	13.8%	—	—	—
Thinnest loc. thickness	29	21	72.4%	0.0%	17.2%	72.4%	82.8%	93.1%
Overall	300	164	54.7%	17.0%	20.3%	61.6% [†]	78.1% [†]	91.7% [†]

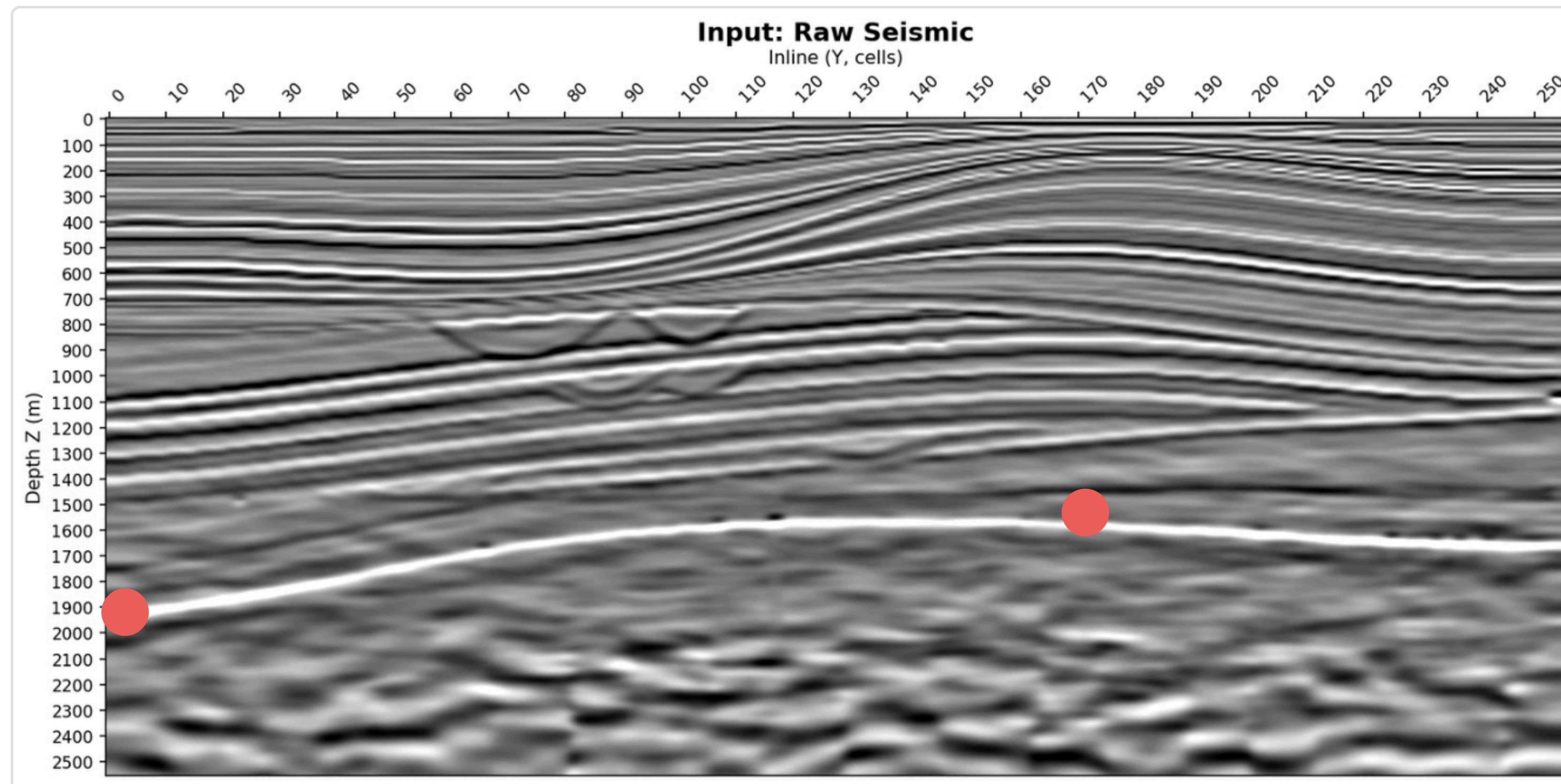
Table 4: Performance by bucket. The rightmost columns report the fraction of thickness predictions within different absolute error tolerances. [†]Computed over all thickness questions ($n = 242$).



Example 1

Identify the lateral location (in image-cell coords) where the sediment is thickest in this seismic image. $GT = 0$

GRAYSCALE AGC PANEL (MODEL INPUT)

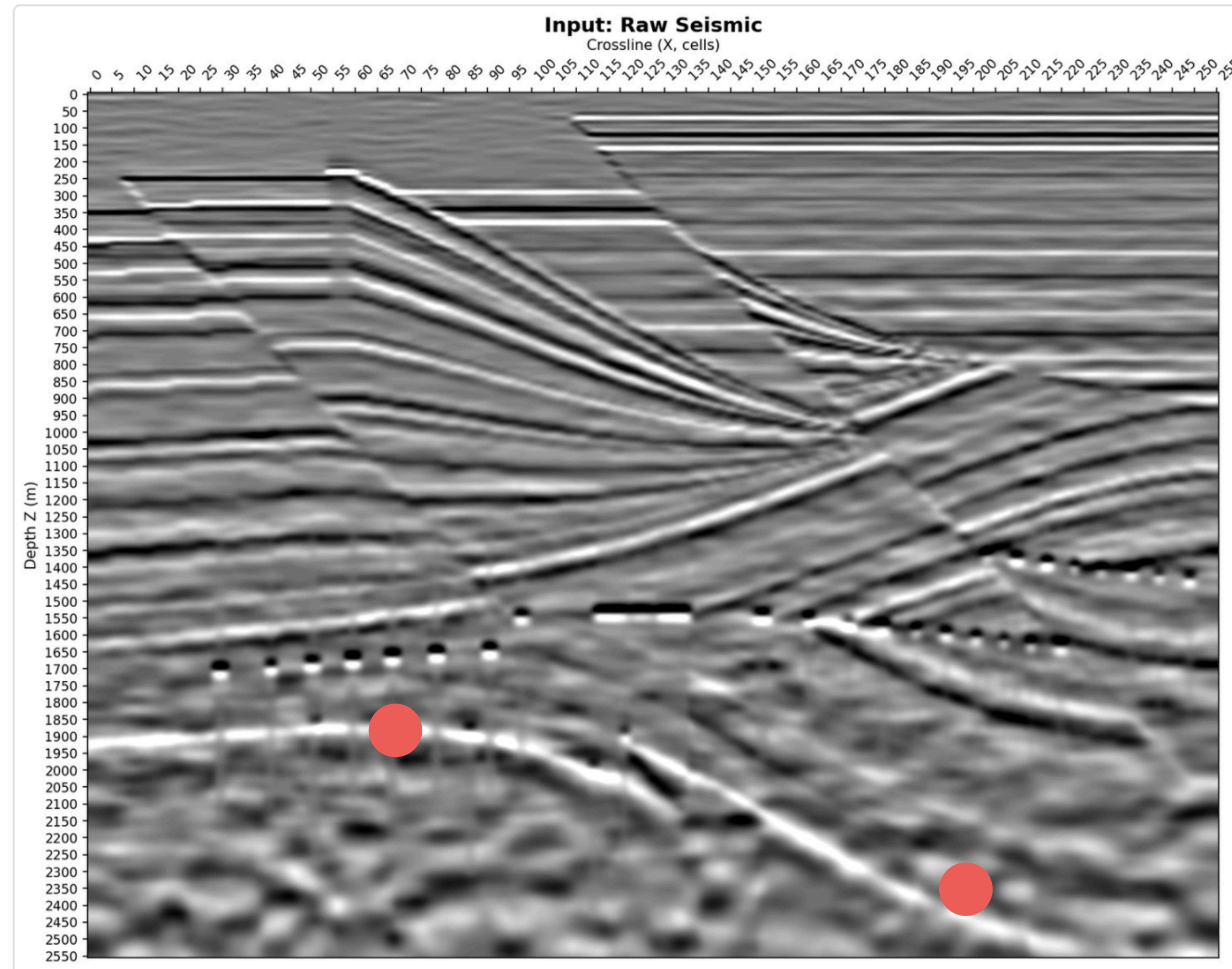


From the seismic display, I obtain each value by scanning a column to the last reflector that holds across adjacent columns and reading its centre-line depth. Reading several columns rather than one guards against a single noisy pick. Reading the deepest coherent horizon at columns 0, 170 gives 1900, 1560 m. That makes column 0 the deepest (1900 m). Rounding column 0 to the nearest 10-cell tick gives 0.



Example 2

Identify the sediment thickness at Crossline (X) = 195 (in cells) in this seismic image, in meters. GT = 2400m



(incorrect)

From the panel, I obtain each value by scanning a column to the last reflector that holds across adjacent columns and reading its centre-line depth. Below this band the reflectors lose coherence, which confirms it is the basement. At columns 64, 195 the horizon sits at 1930, 2370 m. Column Crossline (X) = 195 reads 2370 m. So the thickness is 2370 m.

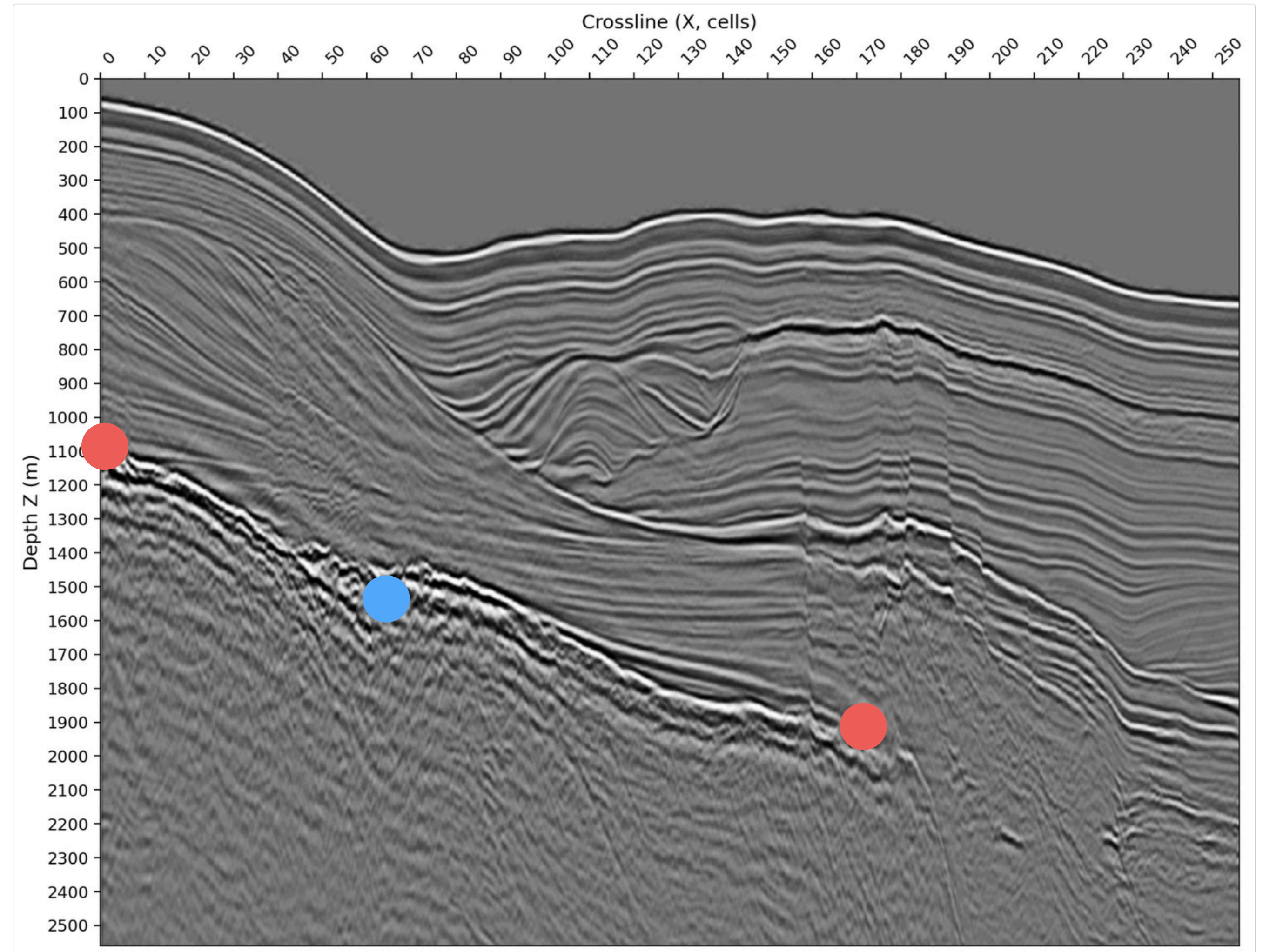


Example 3: Real Seismic Image

[find thickest location]: From the seismic display, I obtain each value by scanning a column to the last reflector that holds across adjacent columns and reading its centre-line depth. Reading several columns rather than one guards against a single noisy pick. The basement depths at columns 0, 170 come out as 1100, 1840 m. That makes column **170** the deepest (**1840 m**). Column 170 rounds to the nearest 10-cell tick, so the answer is 170.

Identify the sediment thickness at Crossline (X) = 64 (in cells) in this seismic image, in meters.

From the panel, I obtain each value by scanning a column to the last reflector that holds across adjacent columns and reading its centre-line depth. Below this band the reflectors lose coherence, which confirms it is the basement. At columns 64, 64 the horizon sits at 1530, 1540 m. Tracing to Crossline (X) = 64, the basement is 1530 m. Answer: 1530 m.



real_world_seismic_test_image_axes.png